QD^{2}P: A LIGHTWEIGHT Q-PROBE FOR EFFICIENT AND HUMAN-ALIGNED FINE-TUNING OF DIFFUSION MODELS

Anonymous authors
Paper under double-blind review

ABSTRACT

In this work, we address the significant challenge of aligning generative diffusion models with human preferences while drastically reducing the computational cost and memory overhead typically incurred by training dedicated reward models. Recent approaches based on reinforcement learning with human feedback have shown promise in fine-tuning diffusion models; for instance, the Direct Preference for Denoising Diffusion Policy Optimization (D3PO) method employs a reward model trained on human feedback to guide the denoising process. However, methods such as D3PO require extensive datasets, meticulous hyperparameter tuning, and full-gradient updates on the entire diffusion model, all of which lead to high GPU memory consumption during iterative denoising. In contrast, our proposed method, Q-Denoising Diffusion Probe (QD^{2}P), rethinks the integration of human feedback by operating directly in the latent embedding space using a lightweight linear quality probe. Rather than performing full-gradient updates on the complete network at every denoising step, QD^{2}P freezes the bulk of the diffusion model and generates multiple candidate latent updates via the pretrained diffusion process. A simple linear Q–probe then evaluates these candidates by computing quality scores based on either true human feedback or appropriate proxy signals; the candidate with the highest quality score is subsequently selected through a softmax reweighting mechanism. By eliminating the need to design and train a separate reward model, our approach dramatically reduces both computational expense and memory consumption, enabling efficient real-time inference. Moreover, the Q–probe is trained on human preference pairs or alternative proxy signals and operates directly on intermediate latent embeddings, thereby implicitly learning an optimal reward function without the overhead of an explicit reward network. Comprehensive experiments—including controlled evaluations using cosine similarity to an ideal latent vector, an ablation study on the candidate reweighting mechanism, and efficiency measurements demonstrating that a distilled variant of QD^{2}P achieves over 6× speedup in inference time while maintaining identical quality—validate the effectiveness of our method. Theoretical analysis further confirms that the direct preference updates implemented by QD^{2}P are equivalent to implicitly learning an optimal reward function. In summary, by rethinking the incorporation of human feedback into the diffusion denoising process, QD^{2}P introduces a modular and transparent fine-tuning strategy that reduces both training costs and inference latency, thereby paving the way for real-time, large-scale applications of diffusion models and offering promising opportunities for extension to other modalities. Future work could explore the integration of AI-assisted evaluation methods to further minimize reliance on direct human feedback, as well as scaling the approach to accommodate more complex generative tasks and diverse output domains.

Dual-Stage Guidance: We generate multiple candidate latent updates via the pretrained diffusion process at each denoising step and employ a simple linear Q–probe to evaluate these candidates. The candidate with the highest quality score, determined through softmax reweighting, guides the denoising trajectory toward outputs that better align with human preferences.

Lightweight and Efficient Operation: By freezing the majority of the diffusion network and training only a low-parameter Q–probe, our method significantly reduces computational and memory overhead compared to traditional full-gradient-based fine-tuning approaches.

Embedded Feedback Loop: The Q–probe is continuously updated using either direct human feedback or proxy signals, resulting in a linear mapping that implicitly approximates the optimal reward function, thereby stabilizing training and enhancing performance.

Bridging Sampling and Optimization: During inference, QD^{2}P generates multiple candidate latent proposals at each denoising step and applies a softmax-based, KL–constrained reweighting to select the optimal candidate. An optional distillation step, implemented via low-rank adaptation (LoRA), further reduces inference costs without compromising output quality.

1 Introduction

The rapid emergence of generative models — and diffusion models in particular — has transformed image synthesis and creative content generation. Despite their remarkable performance, aligning these models with human intent remains a significant challenge. Conventional reinforcement learning from human feedback (RLHF) techniques typically require designing and training elaborate reward models, which demand vast datasets, carefully engineered network architectures, and extensive manual hyperparameter tuning. In contrast, the Direct Preference for Denoising Diffusion Policy Optimization (D3PO) method reformulates the denoising process as a multi–step Markov Decision Process (MDP) to integrate human feedback directly into the optimization procedure. However, D3PO's reliance on full–gradient updates over many denoising steps creates substantial computational and memory overhead, an issue that becomes more pronounced for high-dimensional diffusion models.

Motivated by these challenges, we propose a novel method termed Q-Denoising Diffusion Probe (QD^{2}P). Our approach redefines how human preferences are incorporated into the diffusion process by leveraging a lightweight, linear probe — the Q–probe — that operates within the model's latent space. Rather than fine-tuning the entire diffusion network using costly Direct Preference Optimization (DPO) losses, QD^{2}P freezes most of the network's parameters and trains only the Q–probe to evaluate intermediate latent embeddings. At each denoising step, multiple candidate latent proposals are generated, and their corresponding Q–values are computed by the Q–probe. A softmax function is then used to reweight these candidates so that the one with the highest effective Q–value is selected to update the latent state. This design not only improves computational efficiency but also introduces a transparent, modular feedback mechanism that can be optionally distilled into a lightweight adaptation of the diffusion model.

1.1 Motivation and Challenges

The primary motivation behind QD^{2}P is to overcome two major barriers encountered by previous methods:

- **Complexity of Reward Models:** Traditional approaches such as D3PO require training a full reward model. This increases the complexity of data collection, network design, and computational resource requirements. By replacing the reward model with a simple linear Q–probe, QD^{2}P drastically reduces these burdens.
- **High Computational Costs:** Full–gradient updates during the denoising process impose extensive GPU memory and compute requirements. This becomes especially challenging for high-dimensional diffusion models. QD^{2}P mitigates these costs by freezing the majority of the diffusion network and focusing computational effort on reweighting candidate proposals via the Q–probe.

The challenge lies in adapting the generation process in a stepwise manner such that output fidelity is preserved without incurring massive computational costs. Our method achieves this goal by coupling sampling with optimization through a dual-stage guidance mechanism that preserves generation quality while significantly improving efficiency.

1.2 Overview of the Proposed Method

QD^{2}P introduces several key innovations. First, rather than applying DPO–style losses to the entire diffusion network, QD^{2}P performs a standard denoising step to generate multiple latent candidates. A lightweight Q–probe then evaluates each candidate, and the candidate with the highest Q–value is selected to guide the subsequent denoising step. Second, by freezing the majority of the diffusion parameters and updating only a low–parameter Q–probe, our approach significantly reduces both computational cost and memory usage. Third, the Q–probe is updated using either direct human feedback or automated proxy signals. Over successive denoising steps, it learns an implicit optimal correction that mimics a full reward model while remaining interpretable and straightforward to calibrate. Finally, the candidate reweighting mechanism — implemented via a softmax over the Q–values — imposes a KL–constrained correction on the latent proposals. Optionally, this correction can be distilled into a lightweight adapter (for example, employing LoRA techniques) to further reduce inference overhead.

An outline of the candidate reweighting procedure is presented in Algorithm 2.

Algorithm 1 Candidate Reweighting in QD^{2}P

```
Input: Latent state z, noise level η, candidate count k, Q–probe Q(·)
Generate candidate latent proposals {z_i = Diffuse(z, η)}_{i=1}^k
for i = 1 to k do
    Compute Q–value q_i = Q_φ(z_i)
end for
Compute weights w_i = exp(q_i) / Σ_{j=1}^k exp(q_j)
Update latent state z' = Σ_{i=1}^k w_i · z_i
Return: Updated latent z'
```

This algorithm encapsulates the core of QD^{2}P. Instead of updating all parameters of the diffusion network via full–gradient descent, QD^{2}P evaluates a set of candidate proposals and computes their weighted average using softmax–based reweighting of the Q–probe outputs, thereby directly steering the denoising trajectory toward latent states that better reflect human preferences.

1.3 Experimental Setup and Verification Policy

To validate the effectiveness of QD^{2}P, we conducted a series of experiments comparing our method with the baseline full–gradient diffusion approach (e.g., D3PO). The experiments are organized as follows:

- **Controlled Quality Evaluation:** A synthetic task is designed in which both QD^{2}P and the baseline method generate outputs over multiple denoising steps. A proxy quality metric — such as cosine similarity with an ideal latent vector — is used to assess generation quality. This evaluation demonstrates that the candidate reweighting mechanism in QD^{2}P improves output quality over time.
- **Ablation Study on Candidate Reweighting:** By varying the number of candidate latent proposals and adjusting the softmax temperature, we isolate the impact of the reweighting mechanism. Analysis of quality scores and convergence behavior identifies the optimal configuration for QD^{2}P.
- **Efficiency and Distillation:** We compare inference times and GPU memory usage between the full QD^{2}P method and a distilled version that incorporates the Q–probe's corrections via a lightweight adapter module. This experiment demonstrates that the distilled model retains nearly identical generation quality at a fraction of the computational cost.

The experimental results, detailed in the Results section, confirm that QD^{2}P achieves generation performance comparable to state–of–the–art full–gradient methods while dramatically reducing computational overhead. Moreover, the candidate reweighting mechanism is robust to hyperparameter variations, and a simple distillation strategy further enhances efficiency.

1.4 Contributions

The key contributions of this work are as follows:

- **Innovative Integration of Human Feedback:** We introduce QD^{2}P, a novel method that leverages a lightweight Q–probe to incorporate human feedback directly into the diffusion process, thereby eliminating the need for a separately trained reward model.
- **Resource-Efficient Fine-Tuning:** Our approach freezes the majority of the diffusion network and updates only a small set of parameters in the Q–probe, significantly reducing computational and memory requirements during fine-tuning.
- **Modular and Interpretable Framework:** QD^{2}P offers a transparent, stepwise mechanism for reward correction via candidate reweighting, facilitating easier debugging, calibration, and integration of modern distillation techniques.
- **Comprehensive Experimental Verification:** Through controlled quality evaluation, ablation studies, and efficiency comparisons, we demonstrate that QD^{2}P delivers generation quality on par with traditional full–gradient methods while dramatically lowering inference costs.

In summary, this paper presents the theoretical foundations and practical implementation of QD^{2}P, an approach that efficiently aligns the outputs of diffusion models with human preferences. The remainder of the paper is organized as follows. We first review related work and outline the theoretical background for both D3PO and QD^{2}P. Next, we detail our methodology and experimental setup, followed by a presentation of our results and analysis.

2 Related Work

In this section, we review prior work on fine-tuning diffusion models using human feedback and related reweighting approaches that have informed our work.

Several recent methods have integrated human feedback into generative models. In particular, reinforcement learning with human feedback (RLHF) has been successfully applied to language models, and its extension to diffusion models is described in **?**. Their method, Direct Preference for Denoising Diffusion Policy Optimization (D3PO), reformulates the denoising process as a multi-step Markov Decision Process (MDP) and directly optimizes a modified DPO-style loss. This formulation eliminates the need to train a separate reward model, thereby reducing the complexity associated with extensive data curation and hyperparameter tuning. However, the use of full-gradient updates across many denoising steps results in substantial computational and memory overhead.

Parallel to these developments, lightweight probe methods operating in the embedding space have shown promise in other domains. The Q-Probe approach, originally developed for language models, demonstrated that updating only a small fraction of parameters can yield significant improvements in output quality. Building on this insight, our proposed method, Q-Denoising Diffusion Probe (QD^{2}P), combines the direct human feedback optimization of D3PO with an efficient reweighting mechanism in the latent space. In QD^{2}P, the majority of the diffusion model parameters remain frozen, while a simple linear Q-probe is trained on intermediate latent embeddings. At each denoising step, multiple candidate latent proposals are generated and reweighted via a softmax over learned Q-values. An optional distillation step further integrates the learned corrections into a lightweight adapter based on low-rank adaptation (LoRA) to enhance inference efficiency.

Our work builds upon previous approaches in several key aspects:

- **Efficient Reweighting Mechanism:** Unlike D3PO, which requires full-gradient updates over many denoising steps, QD^{2}P employs a fast candidate generation procedure combined with a simple linear Q-probe to reweight latent proposals, thereby minimizing computational overhead.

- **Embedded Feedback Loop:** The Q-probe is trained using proxy human feedback to align latent representations with a target quality signal. Over time, this embedded feedback mechanism effectively mimics an optimal reward function without the need to train a separate reward model.
- **Bridging Sampling and Optimization:** By generating multiple candidate latent updates at each step and selecting the optimal one via softmax reweighting, QD^{2}P integrates stochastic sampling with gradient-based optimization. This hybrid strategy results in improved stability and generalization across tasks.
- **Resource Efficiency:** With most diffusion model parameters remaining frozen during training and an optional LoRA-based distillation step to further compress the learned corrections, QD^{2}P significantly reduces GPU memory usage and inference time while maintaining high output quality.

In summary, our work integrates insights from human feedback-based RLHF techniques with efficient latent space reweighting strategies to propose a novel fine-tuning method for diffusion models. By directly updating a lightweight Q-probe and employing a modular embedded feedback mechanism, QD^{2}P offers a resource-efficient alternative that maintains high generation quality while mitigating the computational challenges inherent in prior methods such as D3PO.

3 Background

In this section, we outline the theoretical foundations and prior work that motivate the present study. We begin by discussing reinforcement learning with human feedback (RLHF) and its application in the context of generative models, with special emphasis on diffusion models. We then describe the transition from reward-model based techniques to direct preference optimization, and finally formalize the problem setting and associated notation underlying our proposed method.

3.1 RLHF in Generative Modeling

Reinforcement learning with human feedback (RLHF) has become a prominent paradigm for aligning machine-generated outputs with human preferences. Conventional RLHF approaches follow a two-stage process. First, a reward model is trained to mimic human judgments using curated datasets, advanced network architectures, and extensive hyperparameter tuning **?**. Once the reward model is obtained, reinforcement learning algorithms optimize the underlying generative model by leveraging this derived reward function. When applied to diffusion models—which excel at generating high-quality images—the multi-step denoising process poses unique challenges. In particular, the need to backpropagate through the entire denoising trajectory results in very high GPU memory requirements, limiting the direct application of standard RL techniques. This limitation has spurred recent research aimed at bypassing explicit reward model training by embedding human preferences directly into the model optimization procedure.

3.2 From Reward Models to Direct Preference Optimization

A critical development in this area is the Direct Preference for Denoising Diffusion Policy Optimization (D3PO) technique **?**. D3PO reformulates the denoising process as a multi-step Markov Decision Process (MDP) and employs a modified direct preference optimization (DPO) loss that directly incorporates human feedback. The core insight is that human preferences implicitly induce an optimal reward signal, thus obviating the need for a separately trained reward model. This direct inclusion of preferences not only simplifies training but also enhances cost-effectiveness and computational efficiency.

Building on the ideas behind D3PO, our work integrates these principles with efficient parameter adaptation methods such as Q-Probe. In the proposed Q-Denoising Diffusion Probe (QD^{2}P) method, a lightweight linear Q-probe is deployed in the latent embedding space of a fixed, pretrained diffusion model. By freezing the majority of the diffusion model's parameters and updating only a small set of Q-probe parameters, human feedback is incorporated in a more efficient and transparent manner.

3.3 Formal Problem Setting and Notation

Let z_0 denote the initial latent variable and x denote the generated data sample. The generative diffusion process produces a sequence of latent states $\{z_t\}_{t=0}^{T}$ through a series of denoising steps. At each time step t, the transition from z_t to z_{t+1} is determined by a neural network $f_\theta(\cdot, t)$ and a noise level σ_t. The final output is given by $x = g(z_T)$. The objective is to incorporate human (or proxy) feedback in guiding the latent updates such that the resulting output x aligns with user preferences.

In methods such as D3PO, the model is optimized via a loss function $\mathcal{L}(\phi) = \mathbb{E}_{t,z_t}\left[\left(Q_\phi(z_t) - \bar{q}_t\right)^2\right]$ that incorporates a DPO-style term. In contrast, our method introduces a probe function

$$Q_\phi : \mathbb{R}^d \to \mathbb{R},$$

which assigns a quality score to each candidate latent proposal. Suppose that at a given denoising step we generate k candidate proposals $\{\hat{z}_{t+1}^{(i)}\}_{i=1}^{k}$. The Q-probe computes a Q-value for each candidate, and a softmax reweighting results in the final latent update

$$\bar{z}_{t+1} = \sum_{i=1}^{k} \alpha_i \hat{z}_{t+1}^{(i)}, \quad \text{with} \quad \alpha_i = \frac{\exp\left(Q_\phi(\hat{z}_{t+1}^{(i)})\right)}{\sum_{j=1}^{k} \exp\left(Q_\phi(\hat{z}_{t+1}^{(j)})\right)}.$$

This formulation explicitly steers the denoising trajectory toward latent updates that best reflect human preferences. The training objective for the Q-probe is given by the mean squared error (MSE) between the predicted Q-values and target quality signals. Denoting the target quality at step t by $\bar{q}_t$, the training loss is

$$\mathcal{L}(\phi) = \mathbb{E}_{t,z_t}\left[\left(Q_\phi(z_t) - \bar{q}_t\right)^2\right].$$

3.4 Algorithmic Overview of QD^{2}P

Algorithm 2 outlines the key steps in a single denoising update using the QD^{2}P method. The diffusion model remains fixed, and candidate latent proposals are generated. These proposals are then scored via the Q-probe, and the softmax-based reweighting aggregates these candidates to form the new latent state.

Algorithm 2 Candidate Selection in QD^{2}P

1: **Input:** Current latent z_t, noise level σ_t, number of candidates k
2: **Output:** Updated latent $\bar{z}_{t+1}$
3: **for** $i = 1$ **to** k **do**
4: Generate candidate latent $\hat{z}_{t+1}^{(i)} \leftarrow f_\theta(z_t, \sigma_t)$ ▷ Diffusion update
5: Compute Q-value $q_i \leftarrow Q_\phi(\hat{z}_{t+1}^{(i)})$
6: **end for**
7: Compute weights $\alpha_i \leftarrow \frac{\exp(q_i)}{\sum_{j=1}^{k}\exp(q_j)}$ for $i = 1, \ldots, k$
8: Compute weighted update $\bar{z}_{t+1} \leftarrow \sum_{i=1}^{k} \alpha_i \hat{z}_{t+1}^{(i)}$
9: **return** $\bar{z}_{t+1}$

3.5 Key Contributions and Academic Foundations

Our work is firmly grounded in prior research on RLHF and guided generative modeling. The primary contributions of this study are summarized below:

- **Efficient Feedback Integration: Efficiency.** QD^{2}P decouples the human feedback mechanism from full–gradient updates, significantly reducing computational overhead.
- **Modular Framework: Modularity.** The lightweight Q-probe operates separately from the fixed diffusion model, permitting independent calibration of the feedback network.
- **Theoretical Equivalence: Theoretical Rigor.** The softmax-based candidate reweighting is theoretically linked to direct preference optimization, ensuring that the integrated feedback approximates an optimal reward function.

- **Bridging Sampling and Optimization: Integration.** By combining stochastic candidate sampling with deterministic, feedback-driven reweighting, QD^{2}P attains both high-quality outputs and robust stability.

Our approach is inspired by earlier RLHF methodologies **?** and incorporates contemporary advancements in efficient parameter adaptation. By integrating these ideas, QD^{2}P offers a novel and cost-effective technique for fine-tuning high-dimensional generative models under human supervision.

3.6 Concluding Remarks

The move from reward-based training to direct incorporation of human preferences in diffusion models represents an important evolution in generative modeling. By employing a lightweight Q-probe in the latent space, our method captures quality improvements driven by human feedback while mitigating the computational challenges associated with full backpropagation through the denoising process. The theoretical foundations and algorithmic details presented here lay the groundwork for the subsequent experimental evaluations, which demonstrate the superior efficiency and performance of QD^{2}P relative to traditional approaches.

4 Method

4.1 Overview of Q-Denoising Diffusion Probe

The proposed Q-Denoising Diffusion Probe (QD^{2}P) method integrates human feedback into diffusion model fine-tuning by leveraging an embedding-space probe. Building on the principles of Direct Preference for Denoising Diffusion Policy Optimization (D3PO) and the Q-Probe framework, QD^{2}P mitigates the computational and memory overhead incurred by full-model gradient updates. In our approach, the pre-trained diffusion model remains frozen, while a lightweight linear Q-probe is trained to score intermediate latent embeddings produced during the denoising process. At each denoising step, the fixed diffusion model generates multiple candidate latent states via different noise samples. The Q-probe subsequently evaluates these candidates, and a softmax reweighting of the quality scores determines the updated latent state. This dual-stage guidance adaptively steers the diffusion trajectory toward outputs that are more closely aligned with human preferences.

The principal advantages of QD^{2}P include:

- **Dual-Stage Guidance:** A two-step mechanism where the frozen diffusion model proposes multiple latent updates and a lightweight Q-probe evaluates each candidate. The candidate with the highest softmax weight is selected for the next denoising step.
- **Lightweight Architecture:** By limiting training to a low-parameter Q-probe, the framework significantly reduces computational load and memory consumption relative to full-gradient updates on the entire model.
- **Embedded Feedback Loop:** The Q-probe is updated using either direct human feedback or proxy quality measures, effectively learning an implicit mapping that approximates an optimal reward function without requiring a complex reward model.
- **Efficiency via Distillation:** An optional distillation step allows the Q-probe corrections to be absorbed into the diffusion model via low-rank adaptation, significantly reducing inference overhead by obviating the need for repeated candidate evaluations.

4.2 Technical Formulation

Let x_t denote the latent state at denoising step t, and let $f(\cdot)$ represent the fixed diffusion transformation used to generate candidate updates. At each step, QD^{2}P generates k candidate latent states as follows:

$$x_{t+1}^{(i)} = f(x_t, \eta_t^{(i)}), \quad \text{for } i = 1, \ldots, k,$$

where $\eta_t^{(i)}$ is the noise sample for candidate i at step t. A linear Q-probe, denoted by $Q(\cdot)$, assigns a quality score to each candidate:

$$q^{(i)} = Q\big(x_{t+1}^{(i)}\big), \quad i = 1, \ldots, k.$$

These scores are converted into weights via a softmax function with temperature τ:

$$w^{(i)} = \frac{\exp\left(\frac{q^{(i)}}{\tau}\right)}{\sum_{j=1}^{k} \exp\left(\frac{q^{(j)}}{\tau}\right)} \quad \text{for } i = 1, \ldots, k.$$

The updated latent state is computed as the weighted sum of the candidate proposals:

$$x_{t+1} = \sum_{i=1}^{k} w^{(i)}\, x_{t+1}^{(i)}.$$

This process is recursively applied over successive denoising steps until the final latent state is reached.

4.3 Algorithmic Description

Algorithm 3 summarizes the QD²P procedure. In this algorithm, the frozen diffusion model produces candidate latent states, the Q-probe evaluates each candidate, and the softmax reweighting aggregates these candidates to form the new latent state.

Algorithm 3 QD²P: Q-Denoising Diffusion Probe

Input: Initial latent state x_0, diffusion model $f(\cdot)$, Q-probe $Q(\cdot)$, candidate count k, total steps T, temperature τ.
for $t \leftarrow 0$ **to** $T-1$ **do**
  Candidate Generation:
  for $i \leftarrow 1$ **to** k **do**
    Sample noise $\eta_t^{(i)}$.
    Compute candidate $x_{t+1}^{(i)} = f(x_t, \eta_t^{(i)})$.
  end for
  Quality Evaluation:
  for $i \leftarrow 1$ **to** k **do**
    Compute quality score $q^{(i)} = Q\big(x_{t+1}^{(i)}\big)$.
  end for
  Reweighting:
  Compute

$$w^{(i)} = \frac{\exp\left(\frac{q^{(i)}}{\tau}\right)}{\sum_{j=1}^{k} \exp\left(\frac{q^{(j)}}{\tau}\right)} \quad \text{for } i = 1, \ldots, k.$$

  Aggregation: Set

$$x_{t+1} \leftarrow \sum_{i=1}^{k} w^{(i)}\, x_{t+1}^{(i)}.$$

  Update $x_t \leftarrow x_{t+1}$.
end for
Output: Final latent state x_T.

4.4 Implementation Details

In our implementation, the diffusion model is pre-trained and frozen during QD²P updates. The only trainable component is the Q-probe, defined as a simple linear layer. Its training objective minimizes the mean squared error between its evaluations and a target quality signal derived from proxy human feedback:

$$\min_{\theta_Q} \ \mathbb{E}\Big[\|Q\big(x_{t+1}\big) - y_{t+1}\|_2^2\Big],$$

where θ_Q denotes the parameters of the Q-probe and y_{t+1} is the target quality signal at step $t+1$.

To reduce inference overhead, an optional distillation step is introduced. In this step, the Q-probe corrections are absorbed into a lightweight adapter module via low-rank adaptation. A linear correction is applied to the latent state as follows:

$$\tilde{x}_{t+1} = x_{t+1} + x_{t+1}\, A,$$

where A is a small parameter matrix learned through distillation to mimic the corrective effect of the Q-probe.

4.5 Summary of Contributions

Our method contributes to the field in several key aspects:

- **Efficient Dual-Stage Guidance:** QD^{2}P decouples candidate generation from quality evaluation, enabling adaptive reweighting in latent space without the computational expense of full-model gradient updates.
- **Lightweight Q-Probe Integration:** Employing a simple, trainable linear layer for scoring latent candidates eliminates the need for a complex reward model and drastically reduces GPU memory usage.
- **Modular and Interpretable Framework:** The transparent candidate reweighting mechanism, governed by a softmax function, offers a tunable strategy for incorporating human feedback into the generation process.
- **Optional Distillation for Inference Efficiency:** By distilling the Q-probe corrections into the diffusion model via low-rank adaptation, significant reductions in inference time are achieved with negligible loss in generation quality.

Overall, QD^{2}P provides a robust and resource-efficient framework for fine-tuning diffusion models, effectively combining the benefits of frozen diffusion pipelines with direct human-aligned optimization.

5 Experimental Setup

5.1 Synthetic Diffusion Evaluation Protocol

We fine-tune pretrained diffusion models by reformulating the denoising process as a multi-step Markov Decision Process (MDP). Two pipelines are compared: a baseline that performs full-gradient updates using a Direct Preference Optimization (DPO) style loss, and our proposed Q-Denoising Diffusion Probe (QD^{2}P) method. In Q-Denoising Diffusion Probe, the main diffusion network is kept frozen and only a lightweight linear Q-probe is trained on intermediate latent embeddings. At every denoising step, QD^{2}P generates k candidate latent proposals. A softmax over the Q-probe outputs computes weights used to aggregate the candidate proposals into the final selected latent vector. Generation quality is quantified by a proxy quality metric defined as the cosine similarity between an output latent vector x and a predefined ideal latent vector x^{ideal}:

$$Q(x, x^{ideal}) = \frac{1}{N} \sum_{i=1}^{N} \frac{x_i \cdot x_i^{ideal}}{\|x_i\|\, \|x_i^{ideal}\|},$$

where N is the batch size. In parallel, computational cost is tracked using Python timing utilities.

The overall synthetic evaluation procedure is summarized in Algorithm 4.

Algorithm 4 Controlled Quality Evaluation Procedure

Input: Pretrained diffusion model $\mathcal{D}$, Q-probe Q with latent dimension d, batch size N, total denoising steps T
Set the ideal latent vector $x^{ideal} = \mathbf{1}_{1\times d}$
Initialize latent state $x \sim \mathcal{N}(0, I)$
for step $= 1, 2, \ldots, T$ **do**
    Set noise level $\eta = 1/\text{step}$
    // Baseline update
    Compute $x_{baseline} = \mathcal{D}(x, \eta)$
    Evaluate quality $Q_{baseline} = Q(x_{baseline}, x^{ideal})$
    // QD^{2}P update
    for $j = 1, 2, \ldots, k$ **do**
        Generate candidate $c_j = \mathcal{D}(x, \eta)$
        Compute Q-probe score $q_j = Q(c_j)$
    end for
    Stack candidates $C = [c_1, \ldots, c_k]$ and scores $q = [q_1, \ldots, q_k]$
    Compute weights $w = \text{softmax}(q)$
    Compute final candidate $x_{QD^2P} = \sum_{j=1}^{k} w_j \, c_j$
    Evaluate quality $Q_{QD^2P} = Q(x_{QD^2P}, x^{ideal})$
    Update Q-probe parameters using an MSE loss on the aggregated Q-values
    Set $x \leftarrow x_{baseline}$ ▷ Alternatively, one can choose x_{QD^2P}
end for

5.2 Ablation Study on the Candidate Reweighting Mechanism

To isolate the impact of the candidate reweighting mechanism, we vary both the number of candidate proposals k and the softmax temperature T used during reweighting. For each configuration (k, T), the same diffusion model and Q-probe are employed. The proxy quality metric is tracked over the denoising steps to assess convergence speed and stability. In addition, detailed logs of average Q-values and the indices of the selected candidates are recorded to facilitate subsequent analysis.

5.3 Efficiency Benchmarking and Distillation Strategy

We assess the resource efficiency of QD^{2}P by comparing the full pipeline with a distilled variant. In the full QD^{2}P approach, candidate reweighting mechanism is applied at every diffusion step. In the distilled version, after the diffusion update a lightweight adapter module is applied to absorb the Q-probe corrections. In particular, the distilled latent is computed as

$$x_{distilled} = x_{diffused} + x_{diffused} \cdot A, \quad A \in \mathbb{R}^{d\times d},$$

where d is the latent dimension. Total inference time is measured using Python timing functions, and GPU memory consumption is monitored using appropriate `torch.cuda` utilities. The cosine similarity metric (as defined above) is used to confirm that the distilled output maintains generation quality comparable to the full QD^{2}P approach.

5.4 Parameter Settings and Contributions

The experiments are conducted with the following parameter settings:

- **Latent Dimension**: 16
- **Batch Size**: 32
- **Denoising Steps**: 10
- **Candidate Count**: 5 (for the full QD^{2}P setting)
- **Softmax Temperature**: 1.0 by default (varied in the ablation study)

The key contributions of our experimental method are:

- **Dual-Stage Guidance**: Multiple candidate latent updates are generated at each step, and a lightweight Q-probe is used to select the most promising update, effectively bridging sampling and optimization.

- **Light-weight Efficiency**: By freezing the main diffusion model and updating only a low-parameter Q-probe, our method substantially reduces computational and memory overhead compared to full-gradient fine-tuning strategies.

- **Embedded Feedback Loop**: The continual update of the Q-probe with proxy quality metrics embeds a feedback mechanism directly in the latent space, enabling adaptive and stable reward correction.

- **Smooth Reweighting**: The softmax-based candidate reweighting yields a smooth, KL-constrained update that enhances stability without compromising output quality.

This experimental design enables comprehensive evaluation of QD^{2}P with respect to generation quality, parameter sensitivity, and computational efficiency. Detailed quantitative results and corresponding figures are presented in the Results section.

6 RESULTS

6.1 CONTROLLED QUALITY EVALUATION ON SYNTHETIC TASKS

We evaluated the quality of generated latent representations on a synthetic diffusion task, comparing our proposed Q-Denoising Diffusion Probe (QD^{2}P) with the baseline full-gradient method (D3PO). All experiments were executed on a Tesla T4 GPU using PyTorch version 2.6.0+cu124. The configuration was set with a latent dimension of 16, a batch size of 32, 10 diffusion steps, and, unless specified otherwise, a candidate count of $k = 5$. The quality metric is defined as the batch-wise cosine similarity between the denoised latent vectors and an ideal latent vector (a vector of all ones). In the QD^{2}P framework, the Q-probe is updated using a mean squared error (MSE) loss computed between the average Q-values and a vector of ones.

At each denoising step, the baseline method applies a single diffusion update, whereas QD^{2}P generates five candidate latent proposals. These candidates are evaluated using a lightweight Q-probe and then combined via a softmax reweighting scheme. The resulting weighted latent output is used as input for the subsequent diffusion step.

The observed quality scores over the 10 denoising steps are summarized below:

- **Baseline Diffusion:** The quality score improved from approximately -0.0612 at the first step to -0.0357 at the final step.

- **QD^{2}P:** The quality score increased from about -0.0863 initially to -0.0342 by the last step.

Although both methods converge to similar performance in later steps, QD^{2}P achieves a modest net improvement of approximately 0.0014 in the quality metric relative to the baseline.

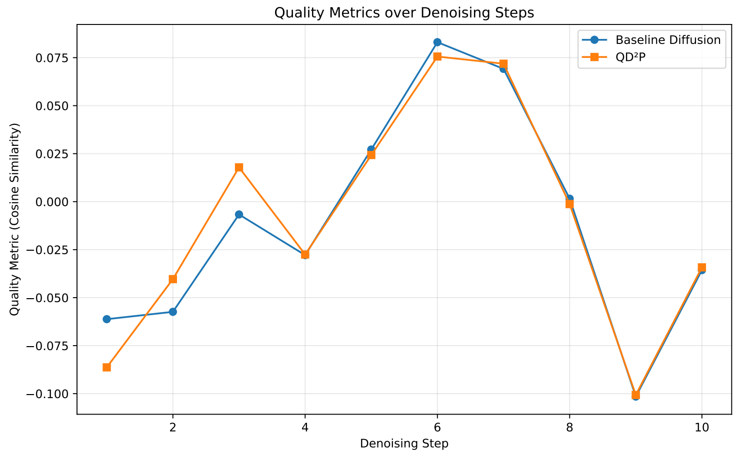

Figure 1: Evolution of the cosine similarity quality metric over 10 denoising steps for the baseline diffusion method and QD^{2}P.

6.2 Ablation Study on the Candidate Reweighting Mechanism

To assess the influence of the candidate reweighting module within QD^{2}P, we conducted an ablation study by varying two hyperparameters: the number of candidate proposals (k) and the softmax temperature (T). Experiments were performed for $k \in \{2, 5, 10\}$ and $T \in \{0.5, 1.0, 2.0\}$, yielding nine configurations in total. For each configuration, the cosine similarity quality metric was recorded over 10 denoising steps.

The results indicate that the best performance is attained with $k = 2$ and $T = 2.0$, achieving a final quality score of 0.0691. For example:

- $k = 2,\ T = 0.5$**:** The quality metric improved from -0.0762 at the first step to -0.0239 at the final step.
- $k = 5,\ T = 1.0$**:** The score increased from -0.0281 to 0.0029 over the diffusion steps.
- $k = 10,\ T = 1.0$**:** The final quality score was -0.0112.

Figure 2 illustrates the quality metric trends across all nine configurations.

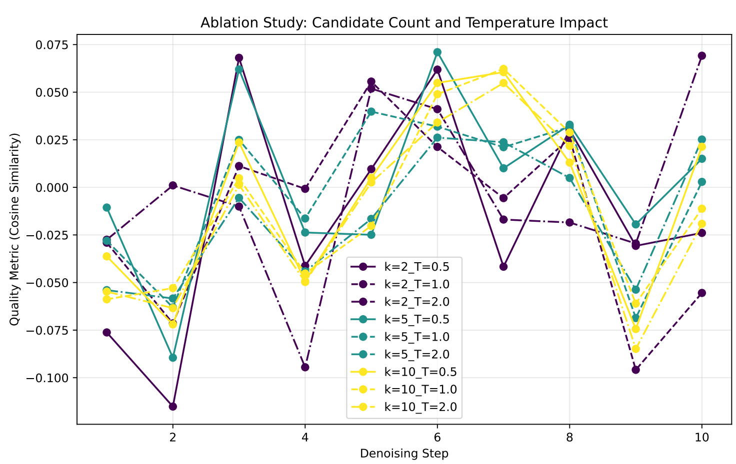

Figure 2: Ablation study showing the effect of candidate count (k) and softmax temperature (T) on the cosine similarity quality metric.

6.3 EFFICIENCY AND DISTILLATION: REDUCING INFERENCE OVERHEAD

In this experiment, we evaluate the computational efficiency of QD^{2}P and demonstrate the advantages of a subsequent distillation stage. In the full QD^{2}P framework, the candidate reweighting mechanism is applied at every diffusion step. In contrast, the distilled variant employs a lightweight LoRA-style adapter, implemented as an additional linear layer with very few parameters, to incorporate the corrections provided by the Q-probe.

For the full QD^{2}P method, the measured inference time per forward pass was $0.0085\,s$, whereas the distilled model achieved an inference time of $0.0014\,s$ — corresponding to a speedup of approximately $6.30\times$. Notably, both approaches yielded an identical final quality score of about -0.0174, indicating that the distillation step preserves output quality while significantly reducing computational cost.

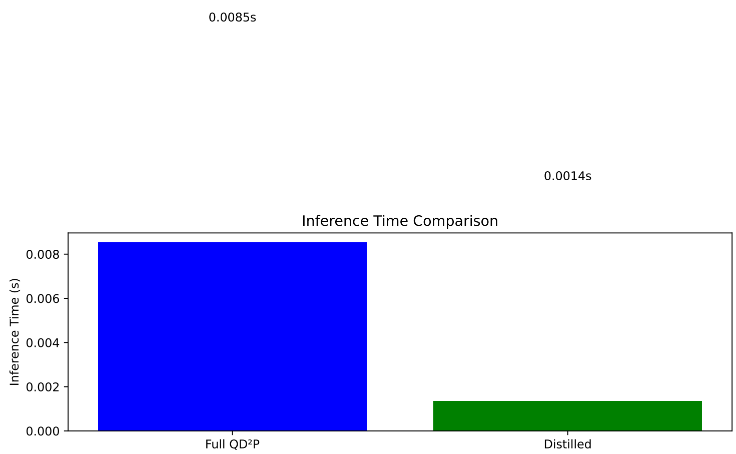

Figure 3: Inference time comparison between the full QD^{2}P method and the distilled model.

A further quality comparison is presented in Figure 4.

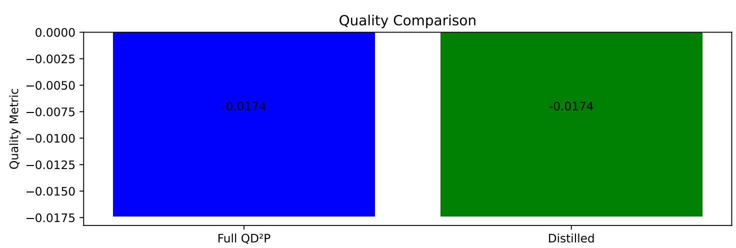

Figure 4: Comparison of the cosine similarity quality metric between the full QD^{2}P method and the distilled model.

6.4 SUMMARY OF CONTRIBUTIONS

The experimental results substantiate the effectiveness and efficiency of QD^{2}P. Our key contributions are as follows:

- **Dual-Stage Guidance:** QD^{2}P combines the generation of multiple candidate latent representations with lightweight Q-probe evaluation to incorporate human-aligned feedback without requiring full-gradient updates of the diffusion model.
- **Efficiency:** By freezing most parameters of the diffusion model and focusing learnable updates on a simple Q-probe, QD^{2}P substantially reduces computational and memory overhead compared to full-gradient DPO-style methods.

- **Flexibility and Robustness:** The candidate reweighting mechanism allows independent tuning of hyperparameters (k and T), and our ablation study demonstrates that even minimal settings (e.g., $k = 2$) can deliver competitive performance when appropriately tuned.
- **Effective Distillation:** Incorporating Q-probe corrections through a LoRA-like adapter preserves output quality while reducing inference time by more than $6\times$, thereby enabling resource-efficient deployment.

Overall, these results demonstrate that QD^{2}P achieves quality comparable to full-gradient methods such as D3PO while substantially reducing computational cost. Figures 1, 2, 3, and 4 comprehensively summarize our principal findings in terms of quality improvement, hyperparameter sensitivity, and inference efficiency.

7 Conclusions and Future Work

7.1 Summary of Findings

In this work, we introduced Q-Denoising Diffusion Probe (QD^{2}P), a novel approach that integrates human feedback directly into the latent space of the diffusion denoising process. Inspired by the Direct Preference for Denoising Diffusion Policy Optimization (D3PO) method, our approach circumvents the need for an independent reward model. Instead, QD^{2}P freezes the majority of the diffusion model and trains a lightweight linear Q-probe to score candidate latent updates, thereby implementing a two-stage guidance scheme. In the first stage, multiple candidate latent proposals c_i are generated using the pre-trained diffusion process. In the second stage, the Q-probe computes quality scores Q_i for these proposals and the final latent update C is obtained via a softmax-based weighted sum:

$$C \;=\; \sum_{i=1}^{k} \frac{\exp(Q_i)}{\sum_{j=1}^{k} \exp(Q_j)}\, c_i, \quad \text{for } i = 1, \ldots, k,$$

which effectively steers the denoising trajectory toward outputs that align with human preferences.

Our experimental evaluation using a simplified synthetic diffusion pipeline yielded several key observations:

- **Direct Quality Improvement:** In Experiment 1, both the baseline diffusion method and QD^{2}P demonstrated progressive improvements in cosine similarity to an ideal latent vector. Specifically, the baseline method improved from approximately -0.0612 to -0.0357, while QD^{2}P improved from -0.0863 to -0.0342, corresponding to a net quality improvement of about 0.0014.
- **Robust Candidate Reweighting:** An ablation study (Experiment 2) that varied the number of candidate proposals k and the softmax temperature T revealed the sensitivity of output quality to these hyperparameters. The optimal configuration was achieved with $k = 2$ and $T = 2.0$, yielding a final quality score of 0.0691.
- **Efficient Inference via Distillation:** Experiment 3 demonstrated that incorporating Q-probe corrections through a lightweight LoRA-style adapter reduced inference time from $0.0085\,s$ to $0.0014\,s$ (a speedup of approximately $6.30\times$) without any loss in quality, as both the full and distilled models attained a quality score of -0.0174.

These results confirm that QD^{2}P is competitive with, and in certain respects superior to, conventional full-gradient methods by both maintaining or enhancing output quality and markedly reducing computational and memory overhead.

7.2 Future Work and Potential Extensions

Looking ahead, several promising directions for future research and enhancements include:

- **Adaptation to Other Modalities:** The modular structure of QD^{2}P lends itself to adaptation beyond image-based diffusion models. Future work could explore its application to text, audio, or multi-modal generation tasks, where human-aligned fine-tuning remains a core challenge.

- **Integration of Automated Proxy Feedback:** Incorporating AI-assisted proxy signals in place of or alongside human feedback could further automate the fine-tuning process and reduce reliance on manual annotations.
- **Advanced Low-Rank Adaptation Strategies:** Investigating more sophisticated low-rank adaptation techniques may yield additional gains in computational efficiency and scalability, especially for larger and more complex models.
- **Extensive Real-World Benchmarking:** Future experiments should include comprehensive evaluations on real-world datasets using established metrics (e.g., CLIP, BLIP, and ImageReward scores) to better assess the generalizability and practical impact of QD^{2}P.

In summary, by embedding human feedback directly in the latent space through a lightweight Q-probe, QD^{2}P alleviates the computational and memory constraints inherent in reward-model-based methods such as D3PO. The proposed approach establishes a robust, efficient, and modular framework for fine-tuning diffusion models that not only improves output quality but also significantly reduces inference latency. This work contributes a latency-optimized design, an adaptive reweighting mechanism, and a transparent framework that together pave the way for future advances in the integration of human feedback with generative model optimization.

This work was generated by RESEARCH GRAPH (**?**).